| Ex. No: 12 | BUILD WEB APP WITH FLUTTER |
|---|---|
| Date of Exercise | 29.09.2026 |
| Git-Hub Host Link | |

**Aim:**

To develop, run, and build a responsive Flutter application for the Web using Flutter and Chrome.

**Description:**

This experiment demonstrates how a Flutter allows developers to create web applications using the same Flutter framework used for mobile applications. In this experiment, a Flutter Web application is created and executed in Chrome. A responsive layout is implemented using LayoutBuilder so that the interface changes based on the browser window width. Finally, a production web build is generated using flutter build web.

**Procedure:**

**Step 1:** Check the Flutter installation and development environment using: flutter doctor

**Step 2:** Check the available devices and verify that Chrome or another web browser is available using: flutter devices

**Step 3:** Create a new Flutter web application using: flutter create flutter_web_app

**Step 4:** Navigate to the application directory using: cd flutter_web_app

**Step 5:** Open the Flutter project in VS Code using: code .

**Step 6:** Create the web interface in lib/main.dart using Scaffold, AppBar, and ElevatedButton.

**Step 7:** Implement a responsive layout using LayoutBuilder to display different layouts for mobile and desktop screen widths.

**Step 8:** Run the Flutter web application in Chrome using: flutter run -d chrome

**Step 9:** Resize the Chrome browser window and verify that the application changes between the Mobile Web Layout and Desktop Web Layout.

**Step 10:** Create the production web build using: flutter build web and verify that the generated files are available in the build/web/ directory

**Program:**

```
import 'package:flutter/material.dart';

void main() => runApp(const FlutterWebApp());

class FlutterWebApp extends StatelessWidget {
  const FlutterWebApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Web Application',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: const Color(0xFF146C68),
          surface: const Color(0xFFF5F7F4),
        ),
        scaffoldBackgroundColor: const Color(0xFFF5F7F4),
        useMaterial3: true,
      ),
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        backgroundColor: const Color(0xFF146C68),
        foregroundColor: Colors.white,
        title: const Text('Flutter Web Application'),
      ),
      body: LayoutBuilder(
        builder: (context, constraints) {
          final width = constraints.maxWidth;
          final isMobile = width < 600;
          final isTablet = width >= 600 && width < 1100;
          final isDesktop = !isMobile && !isTablet;
          final horizontalPadding = isMobile ? 20.0 : (isTablet ? 40.0 : 64.0);
          final gap = isMobile ? 18.0 : 24.0;
```

```
// Stack panels below desktop width; place them side by side above it.
final panels = <Widget>[
  _InfoPanel(
    label: isMobile
        ? 'MOBILE WEB LAYOUT'
        : isTablet
            ? 'TABLET WEB LAYOUT'
            : 'DESKTOP WEB LAYOUT',
    title: 'One app,\nthree screen sizes.',
    description:
        'Resize the browser to see this page adapt to the space available.',
  ),
  _WelcomePanel(
    onPressed: () {
      ScaffoldMessenger.of(context)
        ..hideCurrentSnackBar()
        ..showSnackBar(
          const SnackBar(content: Text('Welcome to Flutter Web!')),
        );
    },
  ),
];

final content = isDesktop
    ? Row(
        crossAxisAlignment: CrossAxisAlignment.stretch,
        children: [
          Expanded(child: panels[0]),
          SizedBox(width: gap),
          Expanded(child: panels[1]),
        ],
      )
    : Column(
        crossAxisAlignment: CrossAxisAlignment.stretch,
        children: [
          panels[0],
          SizedBox(height: gap),
          panels[1],
        ],
      );

return Padding(
  padding: EdgeInsets.symmetric(
    horizontal: horizontalPadding,
    vertical: isMobile ? 28 : 40,
  ),
```

```
          child: Center(
            child: ConstrainedBox(
              constraints: const BoxConstraints(maxWidth: 1200),
              child: isDesktop
                  ? content
                  : SingleChildScrollView(
                      child: content,
                    ),
            ),
          ),
        );
      },
    ),
  );
 }
}

class _InfoPanel extends StatelessWidget {
  const _InfoPanel({
    required this.label,
    required this.title,
    required this.description,
  });

  final String label;
  final String title;
  final String description;

  @override
  Widget build(BuildContext context) {
    return Container(
      constraints: const BoxConstraints(minHeight: 280),
      padding: const EdgeInsets.all(32),
      decoration: BoxDecoration(
        color: const Color(0xFFDCECE7),
        borderRadius: BorderRadius.circular(8),
      ),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          Text(
            label,
            style: Theme.of(context).textTheme.labelLarge?.copyWith(
                color: const Color(0xFF146C68),
```

```
            fontWeight: FontWeight.w700,
            letterSpacing: 1.2,
          ),
        ),
        const SizedBox(height: 18),
        Text(
          title,
          style: Theme.of(context).textTheme.headlineMedium?.copyWith(
              color: const Color(0xFF143C39),
              fontWeight: FontWeight.w700,
            ),
        ),
        const SizedBox(height: 14),
        Text(
          description,
          style: Theme.of(context).textTheme.bodyLarge?.copyWith(height: 1.5),
        ),
      ],
    ),
   );
  }
}

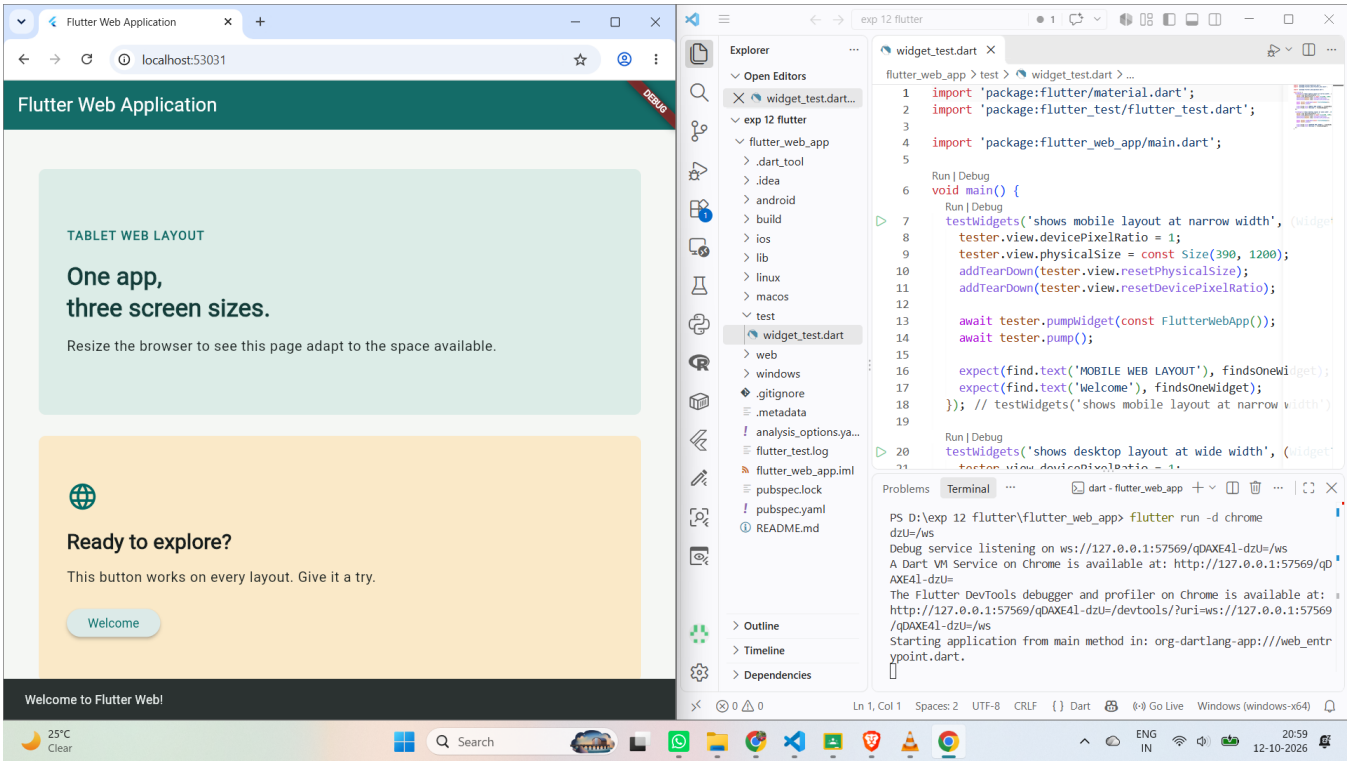
class _WelcomePanel extends StatelessWidget {
  const _WelcomePanel({required this.onPressed});

  final VoidCallback onPressed;

  @override
  Widget build(BuildContext context) {
   return Container(
    constraints: const BoxConstraints(minHeight: 280),
    padding: const EdgeInsets.all(32),
    decoration: BoxDecoration(
     color: const Color(0xFFFAE9C8),
     borderRadius: BorderRadius.circular(8),
    ),
    child: Column(
     crossAxisAlignment: CrossAxisAlignment.start,
     mainAxisAlignment: MainAxisAlignment.center,
     children: [
      Icon(
       Icons.language,
       size: 36,
       color: Theme.of(context).colorScheme.primary,
      ),
```

```
        const SizedBox(height: 18),
        Text(
          'Ready to explore?',
          style: Theme.of(context).textTheme.headlineSmall?.copyWith(
                fontWeight: FontWeight.w700,
              ),
        ),
        const SizedBox(height: 12),
        Text(
          'This button works on every layout. Give it a try.',
          style: Theme.of(context).textTheme.bodyLarge?.copyWith(height: 1.5),
        ),
        const SizedBox(height: 24),
        ElevatedButton(onPressed: onPressed, child: const Text('Welcome')),
      ],
    ),
  );
 }
}
```

## Output:




**RESULT**

A responsive Flutter Web application was successfully developed, executed in Chrome, tested with different screen sizes, and built for production.